Quantum Option Profit and Loss Encoding for Risk Assessment

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***Abstract*—In the financial sector, risk measurement of derivative options has long been a focal point of research. However, traditional risk measurement methods face bottlenecks of low computational efficiency and high complexity. Quantum algorithms, leveraging the properties of quantum superposition and interference, hold promise for enabling parallel exploration of multi-solution spaces in risk assessment, offering more efficient and accurate solutions. Nevertheless, current quantum devices are challenged by noise and errors, posing difficulties in achieving true quantum advantage in the short term. This paper proposes a Quantum Option Profit and Loss Encoding (QOPLE) method for risk analysis, conducting an in-depth study of its application in the risk assessment of European call options, which are widely used in the options market. A Quantum Option Risk Assessment (QORA) model is constructed based on this method. Compared to classical approaches, this model has the potential for quadratic speedup and is more suitable for current NISQ (Noisy Intermediate-Scale Quantum) devices in terms of resource utilization. Finally, we demonstrate the calculation results of risk assessment for European call options using IBM Qiskit. Simulation results show that by appropriately configuring the model's sampling qubits, an accuracy of up to 99% can be achieved, while the number of ancillary qubits required by the model is reduced by at least 15% compared to existing quantum algorithms.**

***Index Terms*—Quantum finance,quantum algorithm, option risk analysis, profit and loss encoding.**

I. Introduction

RISK assessment [1] is a key issue faced by financial institutions because risk indicators help to determine the probability and amount of potential losses in various financial situations, which is crucial to their business [2]. The value at risk (VaR) is commonly used for measuring and controlling financial risk [3], and it has become the mainstream approach for risk management in financial institutions. Financial options are important derivatives for investors to speculate and hedge risks, and the efficiency of option calculation and risk measurement is vital for research and the industry. Inadequate risk measurement and management can lead to institutional bankruptcies, posing a threat to the overall stability of the financial sector. For example, during the 2008 financial crisis, American International Group (AIG) faced huge losses due to improper risk management and high-risk financial derivatives trading. To avoid similar risk events, financial institutions must adopt more precise and efficient risk management tools. This not only requires optimizing existing risk assessment methods but also exploring new technological means to enhance the overall level of risk management.

The Monte Carlo simulation [4], [5] method is one of the commonly used probabilistic and statistical approaches, estimating risk values through random sampling. However, when assessing credit risk, its computational complexity is typically $O(N)$, where N represents the sample size. Although classical methods often employ techniques such as importance sampling to improve performance, it remains difficult to avoid excessive resource consumption [6].

The quantum computing [7] with parallelism and potential exponential acceleration may provide a feasible solution in dealing with above problems [8], [9]. It utilizes the principle of quantum mechanics to process information, paving the way for novel approaches to various computational tasks such as machine learning [10], [11], chemical simulation [12], [13], cryptography [14], [15]. Interestingly, the financial sector is projected to be among the earliest beneficiaries of quantum computing [16]. This is primarily because quantum computing provides a novel approach to solving the problem of many models lacking closed-form solutions in financial theory [6]. Currently, various applications of quantum computing in the financial sector have been proposed. Yao-Hsin Chou et al. [17] proposed a stock assessment strategy based on the trend ratio, which combines improved linear regression and a quantum-inspired tabu search algorithm (ANQTS) to effectively address the issue of the Sharpe ratio overestimating the risk of upward trend portfolios. Tang et al. [18] developed a quantum circuit model for pricing collateralized debt obligations (CDO). Woerner and Egger [19]

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introduced a method using quantum Monte Carlo integration (MCI) to calculate economic capital requirements for financial risk metrics. Orús et al. [20] established a nonlinear network model to predict financial crises. These studies provide new tools and perspectives for the financial industry, demonstrating the potential advantages of quantum computing in addressing traditional financial problems. However, they mainly focus on the pricing of financial instruments and the risk measurement of assets (such as securities and bonds) without attention to derivatives, and ignore the consumption of quantum resources in quantum finance. While derivatives as an important financial application provide investors with more flexible and diversified investment tools definitely, whose risk measurement can enable them to customize risk exposure and investment strategies more carefully. Nevertheless, the risk measurement of derivatives needs to consider more complex factors, such as nonlinear return structure and maturity date, which may lead to more consumption of quantum resources according to the complexity and diversity of the risk measurement. In this situation, we should consider how to maximize the potential of quantum computing in the risk measurement of derivatives with less consumption of quantum resources in the era of NISQ to bring substantial progress to the field of risk assessment. Thus it is expected to propose a more efficient and comprehensive quantum finance solution for risk assessment.

In this study, we focus on solving the risk measurement problem of derivatives with less quantum resource consumption in the NISQ era. We choose the value at risk (VaR) as an indicator of risk measurement, and construct a corresponding quantum risk assessment model for financial derivatives options with non-linear payment structure. Our contributions can be summarized as follows:

- The quantum option profit and loss encoding (QOPLE) algorithm is proposed to address the challenges of handling complex financial derivative options on quantum computers by effectively mapping the profit and loss of options into quantum circuits.
- The quantum option risk assessment (QORA) model for practical risk assessment is constructed based on the QOPLE algorithm. Additionally, the model adopts the maximum likelihood-based quantum amplitude estimation, significantly reducing the consumption of quantum resources.
- The simulation results for assessing the risk of European call options on IBM Qiskit demonstrate that QORA model can still achieve a 99% accuracy even with a reduction of at least 15% in quantum ancillary qubits. It provides a practical example for solving the risk measurement problem of derivatives with less quantum resource consumption in the NISQ era.

The remaining organization of this paper is as follows. Related work and basic concepts are summarized in Section II. In Section III and the Section IV, the QOPLE algorithm and the QORA model of quantum option risk assessment are introduced respectively. In Section V, we design experiments for simulation demonstration and analysis. Finally, the conclusions are drawn in Section VI.

II. Related Work

In this section, we commence with a review of the concept of options and introduce the commonly employed risk measurement methods in classical calculations. Then, we delve into the existing research findings pertaining to quantum risk assessment.

A. Option and Its Risk Measurement

As a leveraged financial instrument, the option itself has a great risk. Further, due to the asymmetric risk and information between the two parties of the option transaction, the measurement of the option risk is more difficult than other financial assets. Therefore, how to accurately measure the option risk and manage the risk has become one of the main research problems of financial risk management.

1) Option Overview: Option is a kind of financial derivative, which refers to the right of buyers and sellers to buy and sell the underlying assets at a certain time in the future at the agreed price [21].

According to the behavior granted to the holder by the option contract to buy or sell the underlying asset, it can be divided into call options and put options. According to the different provisions on the performance time, it can be divided into European options and American options [22]. The former has the right to exercise the right to trade only on the maturity date of the option, while the latter can exercise the right to trade at any time during the trading period.

Given the multitude of option types and the prevalence of a wide range of option styles embedded within various other financial instruments, the accurate assessment and analysis of option risks hold significant importance.

2) Risk Measurement Study of Option: VaR is a commonly used indicator of option risk measurement. It is specifically explained that in the case of changes in the market environment, assets or portfolios may suffer losses [23]. This indicator can be viewed from a holistic perspective rather than being limited to a single sensitive factor. This method can be used to provide new ideas for asset management of future operating institutions. VaR is defined as follows:

$$p(\Delta I > VaR) = 1 - \alpha \tag{1}$$

where ΔI represents the loss value of the asset. VaR is the loss value of the asset ΔI in a changing market environment at the confidence level α or the loss value of the corresponding ΔI at a given quantile.

The key to VaR estimation [24] lies in describing the distribution of profit and loss for an investment during the assessment period [25]. The calculation methods mainly include the following three categories:

Historical simulation method: The historical simulation method starts from the option transaction itself and uses the historical data to derive the actual distribution of the return rate so as to estimate the quantile at a corresponding confidence level, which is the option VaR under the historical simulation method. However, this approach overlooks the dynamic nature

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of the market. If there is a significant difference between the future distribution of market risk factors and their historical distribution, the assessment results would be highly inaccurate.

Parametric method: The parameter method first assumes that the return series of the portfolio obeys a certain distribution and estimates the parameters according to the historical data. According to the assumed distribution and the estimated parameters, the lowest return rate at a given confidence level can be obtained. The value is multiplied by the size of the holding asset position to obtain VaR. Because of its oversimplified nature, this approach does not closely match real-world data, especially when dealing with fat-tailed financial data, making it susceptible to underestimating risks.

Monte Carlo simulation method: The Monte Carlo simulation method first assumes a distribution for the returns, and the validity of this distribution assumption is crucial for the computed results. By estimating parameters for the distribution assumption, future price paths for the underlying asset are simulated, subsequently leading to the calculation of derivative product prices. Finally, these prices are sorted from lowest to highest to identify the αth percentile price, which represents the sought-after VaR. This method basically overcomes the problems encountered in the parameter method and historical simulation method of VaR model. Its main disadvantage is that the calculation is cumbersome, and the computer performance requirements are high, especially the risk measurement with high time requirements.

B. Quantum Risk Assessment

To overcome these limitations, researchers have explored new methods based on quantum computing. By leveraging its unique quantum superposition and interference effects, quantum computing has the potential to offer more efficient solutions for handling complex risk analysis tasks [26]. Among these methods, quantum amplitude estimation (QAE) has shown promise in estimating risk values, offering the potential for quadratic speedup compared to classical Monte Carlo methods [27], [28].

Woerner and Egger [29] proposed a gate-based quantum algorithm for VaR analysis and demonstrated its small-scale implementation on a quantum computer. However, the complexity of the case study was relatively low, and nonlinear payoff structures were not considered, which limits its application in real-world financial scenarios. Building on this, Egger et al. [19] later showcased the quadratic speedup potential for quantum economic capital estimation. They developed a quantum circuit for Gaussian conditionally independent models and validated the model's effectiveness in small-scale simulations. However, the method relies on a single-factor risk model, and the required quantum circuit depth exceeds current hardware capabilities [30].

In recent studies, some efforts have been dedicated to addressing the practical limitations of quantum risk models. The literature [31] presents an improved Credit Risk Analysis (CRA) quantum algorithm, capable of handling more complex default probability models and incorporating non-integer loss parameters, which enhances the model's flexibility and practicality.

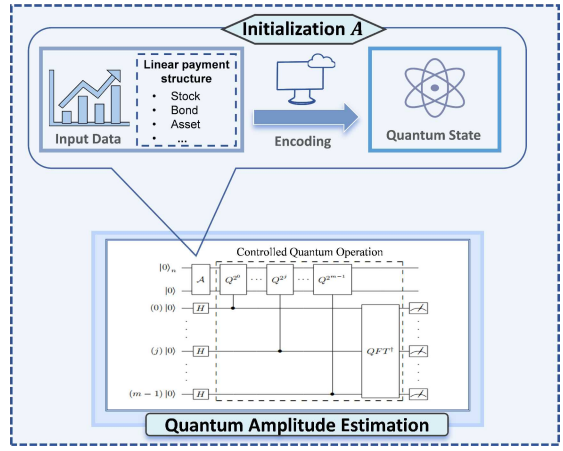

Fig. 1. The existing quantum risk assessment framework.

Although this improvement increases the circuit depth and width, it provides a pathway for more realistic quantum risk management software.

The literature [32] analyzes the practical application requirements of quantum computing in financial risk management, using market risk VaR and potential future exposure (PFE) for credit risk as examples. It discusses the core characteristics and implementation paths of quantum models. The study points out that the current number of qubits and noise levels in quantum hardware remain the primary bottlenecks, and truly practical risk measurement systems will require a several-fold improvement in quantum hardware capabilities.

Existing quantum risk measurement models have shown preliminary effectiveness in measuring the risk of classical financial assets (such as securities and bonds), as illustrated in Fig. 1. The core of the model lies in constructing the Hermitian operator $\mathcal{A}$, which encodes the return distribution structure of financial products into quantum states. The QAE algorithm efficiently estimates the quantum state amplitude through controlled amplification operations and inverse Fourier transforms, providing a quantum acceleration path for calculating metrics such as Value at Risk (VaR). Specifically, suppose that operator $\mathcal{A}$ acts on $n+1$ qubits such that:

$$|\psi\rangle = \mathcal{A}|0\rangle_{n+1} = \sqrt{a}\,|\psi_1\rangle_n\,|1\rangle + \sqrt{1-a}\,|\psi_0\rangle_n\,|0\rangle \tag{2}$$

where $a \in [0,1]$ represents the probability of the interest state, while $|\psi_0\rangle_n$ and $|\psi_1\rangle_n$ are two normalized quantum states. By defining the parameter $\theta_a \in [0, \pi/2]$ to make $sin^2\theta_a = a$, the above equation can be expressed as

$$|\psi\rangle = sin\theta_a\,|\psi_1\rangle_n\,|1\rangle + cos\theta_a\,|\psi_0\rangle_n\,|0\rangle \tag{3}$$

The controlled amplification operation increases the amplitude of the target quantum state through multiple operations. After applying the Q operation m times to $|\psi\rangle$, we obtain the following state:

$$\mathbf{Q}^m|\psi\rangle = \sin\left((2m+1)\theta_a\right)|\psi_1\rangle_n\,|1\rangle + \cos\left((2m+1)\theta_a\right)|\psi_0\rangle_n\,|0\rangle \tag{4}$$

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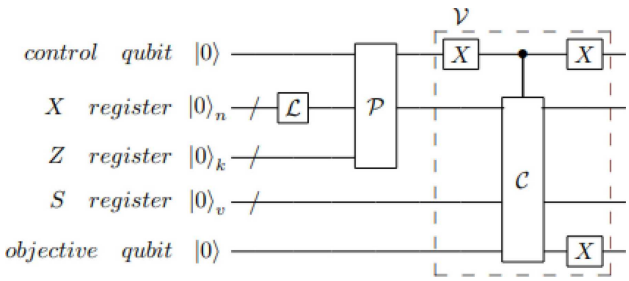

Fig. 2. The framework for quantum option profit and loss encoding can be represented as $\mathcal{A} = \mathcal{VPL}$. First, the operator $\mathcal{L}$ acts on the X register, composed of n quantum bits, to load the option price distribution. Next, the $\mathcal{P}$ operation is used to compute the option's profit and loss distribution. When the option price is less than the strike price K stored in the Z register (with k bits), the control qubit is flipped to the state $|1\rangle$; otherwise, the control qubit remains unchanged. Then, the operation $\mathcal{V}$ (shown by the box in the diagram) is applied to compute the cumulative profit and loss function of the option. This operation compares the option's profit and loss with a given threshold value l stored in the S register (with v bits) using a controlled comparator. If the value is less than or equal to l, the target qubit is flipped; otherwise, the target qubit remains unchanged.

This equation shows that after applying the Q operation m times, for sufficiently small parameter a, the probability of obtaining a good state is at least $4\,\mathrm{m}^2$ times greater than the probability of obtaining the state $|\psi\rangle$, thereby achieving the purpose of probability amplification. Finally, the inverse Fourier transform is applied to convert the phase information of the quantum state into amplitude information, allowing the results obtained through quantum measurement to effectively reflect the amplitude of the quantum state.

However, the applicability of these models is significantly limited in the risk measurement of complex financial instruments, particularly financial derivatives. The return structure of derivatives often exhibits pronounced non-linear characteristics, and their risk exposure is often dependent on multiple market variables and maturities, which places higher modeling demands on traditional quantum risk models [33]. Furthermore, quantum circuits executing high-dimensional, multi-factor computations require deeper circuit structures and more controlled operations, which further exacerbates the limitations of quantum bit numbers and gate depth under NISQ devices. In this context, improving the adaptability of quantum risk models to complex derivatives becomes an urgent issue. This requires not only overcoming the traditional linear mapping methods during operator construction but also optimizing the amplitude estimation process to mitigate the bottlenecks caused by the limitations of quantum hardware resources.

III. Quantum Option Profit and Loss Encoding

In this section, we propose a quantum option profit and loss encoding (QOPLE) algorithm for nonlinear payment structures. The quantum circuit framework is shown in Fig. 2. Its main task is to load the cumulative distribution function (CDF) of options profit and loss (PnL) into the quantum circuit, which mainly involves three operations:

- $\mathcal{L}$, which loads the possible price distribution of expiration date options.
- $\mathcal{P}$, which calculates the profit and loss distribution of the option and divides the option profit and loss into two mapping situations by the auxiliary qubit $|c\rangle$.
- $\mathcal{V}$, which computes the CDF of option profit and loss. This process involves using the quantum state of the target qubit to partition option returns into two categories: those below or equal to a specified level and those above the specified level.

Through the above three operations, the final QOPLE algorithm can be expressed as the operator $\mathcal{A} = \mathcal{VPL}$.

A. Loading Option Price Distribution

The first part of the QOPLE algorithm is to load the option price distribution operation $\mathcal{L}$. Through this operation, the possible price probability distribution of the option at time T is loaded into the quantum register so that a quantum fundamental state corresponds to a possible price, and its amplitude represents the corresponding probability.

The price change of the option follows the BSM model [34], [35], which holds that the value of the option is essentially the cost of replicating the portfolio [36]. In the BSM model, the underlying asset price S_T is related to the time to the expiration time T, the initial price S_0 and other factors, and it follows the lognormal distribution. Since the log-normal distribution has a known analytical expression, it can be efficiently approximated and loaded using quantum circuits such as parameterized rotation gate arrays with polynomial complexity. Therefore, the distribution loading operation $\mathcal{L}$ does not require enumerating all 2^n sample points. We utilize the built-in methods from the uncertainty modeling module in Qiskit, which construct specifically structured quantum circuits to efficiently load the log-normal distribution with polynomial complexity, thus avoiding exponential classical computational overhead.

To encode the price distribution of the option in the quantum state, we truncate the distribution of S_T to the range $[S_{Tmin}, S_{Tmax}]$, where we adopt an interval of three times the standard deviation around the mean. Subsequently, we perform equidistant sampling of S_T within this interval, with a total of 2^n samples, where n represents the number of quantum bits used to store the samples. These samples follow a log-normal distribution, allowing us to calculate the probability of each sample occurrence. In this way, the distribution of S_T can be represented using these n quantum bits. The sampled values of S_T correspond to 2^n numbers represented by n quantum bits. The probabilities of the samples are stored in this set of quantum bits as the squares of the amplitudes of these numbers. In other words, corresponding to an n-qubit quantum register, the operation $\mathcal{L}$ is defined as follows:

$$|0\rangle_n \xRightarrow{\mathcal{L}} |\varphi\rangle_n = \sum_{i=0}^{2^n-1} \sqrt{p_i}|i\rangle_n \tag{5}$$

where $i \in \{0, 1, \ldots, 2^n - 1\}$ denotes the integer in a binary representation corresponding to the possible prices S_T of the

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options at expiration, $|i\rangle_n$ is the quantum state corresponding to each price, and p_i represents the associated probability.

B. Calculate the Option Profit and Loss

Due to the difference between rights and obligations, leverage effect and other factors, the value of the option has a complex and nonlinear response to the change of the underlying asset price. Compared with other financial trading methods, the profit and loss of option trading have nonlinear characteristics, and complex profit and loss analysis is needed to fully understand the risks and returns of option trading. The calculation of the profit and loss of options is usually based on the difference between the market value of the option and the cost of the option, including margin during the option trading process, premiums used to purchase the option, and other fees. Although this should be considered in actual calculations, we are currently temporarily ignoring this part for the purpose of simplifying the model. We focus on the calculation of the intrinsic value and time value of the option.

Assuming K represents the exercise price specified in the options contract, the PnL of the option, denoted as P_T, can be expressed as:

$$P_T = \begin{cases} \max(S_T - K, 0), & \text{call option} \\ \max(K - S_T, 0), & \text{put option} \end{cases} \tag{6}$$

Take call options as an example. When $S_T < K$, the option buyer forgoes exercising the option and does not acquire the underlying asset, resulting in a PnL of 0. When $S_T = K$, the execution or non-execution of the option does not affect the PnL for the option buyer, which remains at 0. When $S_T > K$, the option buyer will choose to exercise the option and acquire the underlying asset, resulting in a profit of $S_T - K$. The option buyer's income is exactly the seller's loss, and the corresponding option buyer's loss is the seller's income. In theory, the maximum loss of the option buyer is limited, but the profit is unlimited, which means that the risk that the option seller needs to bear may be unlimited, which is also the significance of the risk assessment of the option. By evaluating the profit outcomes for the option buyer at the α level, we can directly correspond to the risk assessment for the option seller at the same α level.

We use the operation $\mathcal{P}$ to represent this maximum operation as a quantum circuit. Add the auxiliary bit $|c\rangle$, and compare the option price S_T with the option's exercise price K through the quantum comparator [37]. The operation $\mathcal{P}$ flips the auxiliary qubit $|c\rangle$ from the $|0\rangle$ to the $|1\rangle$ if the option price is lower than the strike price; otherwise, it leaves the auxiliary qubit in the $|0\rangle$. The definition of operation $\mathcal{P}$ is

$$\begin{aligned} |\varphi\rangle_n|0\rangle &= \sum_{i=0}^{2^n-1} \sqrt{p_i}|i\rangle_n|0\rangle \\ &\xRightarrow{\mathcal{P}} |\phi\rangle_{n+1} = \sum_{S_T \le K} \sqrt{p_i}|i\rangle_n|1\rangle + \sum_{S_T > K} \sqrt{p_i}|i\rangle_n|0\rangle \end{aligned} \tag{7}$$

The $\mathcal{P}$ operation divides the profit and loss situation into two categories through the auxiliary bit: the PnL is 0, and the PnL is greater than 0. According to the state of the auxiliary bit, we apply different forms of mapping: the state of the auxiliary bit $|1\rangle$ is mapped to the profit and loss value of 0, and the state of the auxiliary bit $|0\rangle$ is mapped to the corresponding profit and loss value. That is, the quantum state generated by manipulating $\mathcal{P}$ can be expressed as:

$$|\phi\rangle_{n+1} = \sum_{P_T \le 0} \sqrt{p_i}|i\rangle_n|1\rangle + \sum_{P_T > 0} \sqrt{p_i}|i\rangle_n|0\rangle \tag{8}$$

C. Compute the Cumulative Profit and Loss of Option

Even though we have obtained the probability distribution of individual values for option returns, we also aspire to gain a deeper understanding of the cumulative probability distribution across the entire range of returns. This is essential for depicting the probability of option returns reaching or surpassing specific values, thereby providing a more comprehensive insight into the distribution of option returns. To achieve this objective, we have devised the operator $\mathcal{V}$ for computing the cumulative distribution function of option returns.

The main function of operating $\mathcal{V}$ is to sum the profit and loss distribution function to obtain the CDF, which can be expressed as $\mathbb{P}[P_T \le l]$, where l represents the profit and loss level. That is, the cumulative distribution less than l is constructed according to the option profit and loss calculated by operating $\mathcal{P}$.

Therefore, the calculation idea of operating $\mathcal{V}$ is to give a non-zero threshold l and use the controlled comparison circuit to control the profit and loss value P_T to compare with the given l, where the control bit is the auxiliary bit $|c\rangle$. When $|c\rangle$ is $|1\rangle$, it means that the PnL of the quantum state corresponding to 0 will be less than l, so there is no need to compare and flip the target qubit directly to the $|1\rangle$. When $|c\rangle$ is $|0\rangle$, the control comparison circuit compares the PnL with the given l. If the PnL is less than l, the target qubit is flipped from $|0\rangle$ to $|1\rangle$; otherwise, the target qubit is kept as $|0\rangle$. The definition of the final operation $\mathcal{V}$ can be expressed as

$$\begin{aligned} |\phi\rangle_{n+1}|0\rangle &= \sum_{P_T \le 0} \sqrt{p_i}|i\rangle_n|1\rangle|0\rangle + \sum_{P_T > 0} \sqrt{p_i}|i\rangle_n|0\rangle|0\rangle \\ &\xRightarrow{\mathcal{V}} \sum_{P_T \le 0} \sqrt{p_i}|i\rangle_n|1\rangle|1\rangle + \sum_{0 < P_T \le l} \sqrt{p_i}|i\rangle_n|0\rangle|1\rangle \\ &\quad + \sum_{l < P_T} \sqrt{p_i}|i\rangle_n|0\rangle|0\rangle \end{aligned} \tag{9}$$

The probability that the target qubit is $|1\rangle$ corresponds to the probability that the profit and loss value is less than l, that is, $\mathbb{P}[P_T \le l]$ [29]. By adjusting the value of the profit and loss level l, the CDF of the option profit and loss under various values can be obtained. This cumulative probability actually represents the probability that the option profit and loss is less than a specific value. This information is crucial for decision-makers and risk managers in the financial sector. It not only helps us understand the risks of options and portfolios but also has a broad impact on key areas such as decision-making, risk management, and asset pricing.

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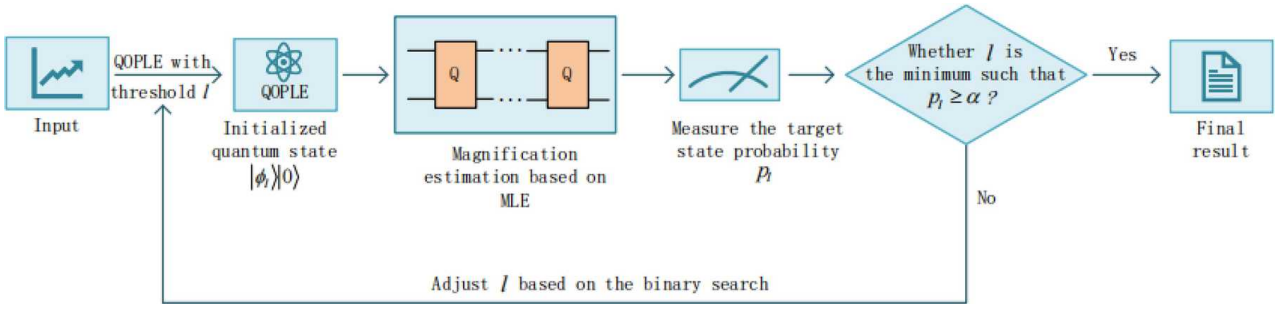

Fig. 3. Quantum option risk assessment model. We use the QOPLE algorithm to construct the quantum state $|\phi_l\rangle|0\rangle$ of the cumulative profit and loss distribution of the option under the specified threshold l. Then, the maximum likelihood estimation (MLE) is used to amplify the quantum state, and then the target state is measured. The value of l is adjusted according to the binary search until the minimum l is found to satisfy $p_l \geq \alpha$, which is the VaR of the option.

IV. Quantum Option Risk Assessment Model

In this section, we propose a quantum option risk assessment (QORA) model based on QOPLE algorithm and deeply study how the QOPLE algorithm is applied to risk assessment.

The overall architecture of the QORA model is shown in Fig. 3. We select VaR as the standard to measure the risk of options. According to the QOPLE algorithm, the option profit and loss distribution function is initialized into the form of quantum state, and the amplitude estimation method based on maximum likelihood estimation is combined with the binary search algorithm to finally determine the risk value of the option at the specified confidence level α.

A. Maximum Likelihood Estimation Method for Estimating Risk Value

For the constructed quantum state $|\phi\rangle_{n+1}|0\rangle$, it is necessary to estimate the probability that the target qubit is $|1\rangle$ state for subsequent operations.

Here, we employ an amplitude estimation method based on Maximum Likelihood Estimation (MLE) [38] to estimate the amplitude of the target quantum state. This method consists of two parts: amplitude amplification and maximum likelihood estimation. Amplitude amplification enhances the probability of the target state through repeated operations. Each amplitude amplification operation increases the probability of the target qubit being in the $|1\rangle$ state. By applying amplitude amplification to the target qubit a different number of times, we measure the number of times the target state is found to be $|1\rangle$ and construct the sample data. Then, the Maximum Likelihood Estimation method is used to identify the parameter that maximizes the probability of occurrence in the sample [39].

We first apply different numbers of amplitude amplification operations to the target qubit based on the set $\{f_q\} = \{f_0, f_1, \ldots, f_R\}$, where the operations are implemented using the Grover operator. After each amplitude amplification, we perform N_q measurements on the amplified target qubit to obtain the number of times the target state is $|1\rangle$ for each amplification count, denoted as $\{h_q\} = \{h_0, h_1, \ldots, h_R\}$, which represents the sample data. These measurement results are used to construct a likelihood function.

$$L(\mathbf{h};\theta_a) = \prod_{q=0}^{R} L_q(h_q;\theta_a) = \prod_{q=0}^{R} \left[\sin^2((2f_q+1)\theta_a)\right]^{h_q} \left[\cos^2((2f_q+1)\theta_a)\right]^{N_q-h_q} \tag{10}$$

where $\mathbf{h} = (h_0, h_1, \ldots, h_R)$. By employing maximum likelihood estimation, we calculate the value of θ that maximizes this particular likelihood function $L(\mathbf{h};\theta_a)$, thereby obtaining an estimate for the amplitude a:

$$\hat{\theta}_a = \arg\max_{\theta_a} L(\mathbf{h};\theta_a) = \arg\max_{\theta_a} \ln L(\mathbf{h};\theta_a). \tag{11}$$

The selection of $\{f_q\}$ is based on the classical follow-up treatment, which employs the Carmer-Rao formula in statistics and the Fisher information measure to evaluate the validity of the parameter estimates and adjust the design of series accordingly. We use an exponentially increasing sequence as the amplification sequence, where the increments are as follows: $f_0 = 0$, $f_1 = 2^0$,..., until $f_m = 2^{m-1}$, and the amplification sequence selection is referred to [39].

This amplitude estimation method based on maximum likelihood estimation estimates the unknown parameters with the maximum likelihood principle, which reduces the controlled operation required by the traditional QAE algorithm [40] and the complex inverse quantum Fourier transform while retaining the advantage of quadratic acceleration, which has more advantages in the face of NISQ device limitations. The final quantum option risk assessment circuit framework based on maximum likelihood estimation is shown in Fig. 4.

B. Binary Search Option VaR

It can be seen from Section II-A2 that the VaR value of the asset is to calculate the minimum loss value greater than the given α confidence level. For the cumulative profit and loss distribution of the option constructed by us, that is, the calculation satisfies $\mathbb{P}[P_T \leq l] \geq \alpha$, that is, the corresponding VaR [26].

In the QOPLE algorithm, we use the operation $\mathcal{V}$ to divide the profit and loss of the option into the target part and the non-target

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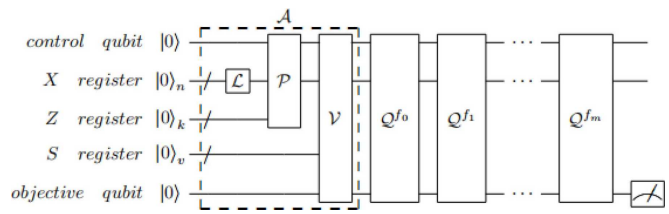

Fig. 4. The circuit framework diagram for quantum option risk assessment based on maximum likelihood estimation. Apply different numbers of amplitude amplification operations to the target qubit, where each amplification corresponds to $\{f_q\}$ times.

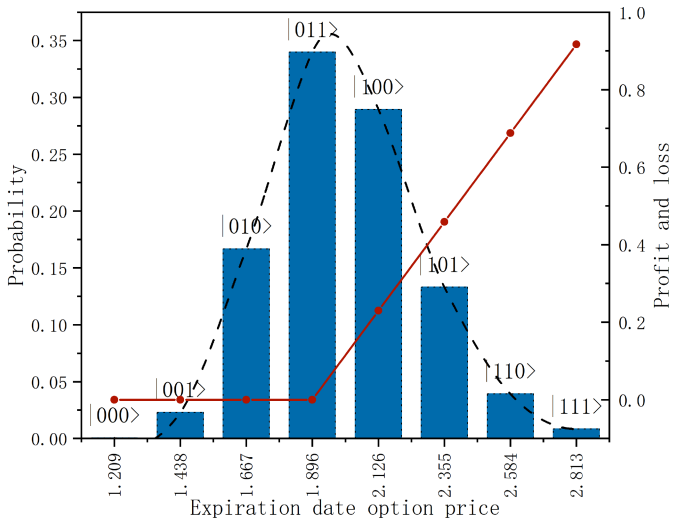

Fig. 5. The distribution of the European call option is loaded with three qubits. The bar chart represents the discrete probabilities associated with the corresponding option prices, and they are labeled with the quantum states to which they have been loaded. The black dotted line corresponds to the original lognormal distribution of the option price S_T in the BSM model. The red solid line represents the corresponding profit and loss of the option price S_T when the exercise price is $K = 1.896$.

part according to the profit and loss level l, where the target part refers to the part whose profit and loss value is less than the given level l. Then, the maximum likelihood estimation algorithm is used to measure and estimate the target part, so as to obtain the cumulative probability corresponding to the specific income level, that is, the probability that the option profit and loss is less than the specific value. By judging the relationship between this probability value and α, combined with the binary search method to continuously adjust l, the risk value of the option can be obtained.

During the binary search process, to ensure that the search range covers the entire probability distribution, we set the upper and lower bounds to the maximum and minimum values of the profit and loss, respectively. Therefore, the computational complexity of the binary search is $O(\log_2(\frac{l_{\max}-l_{\min}}{\epsilon}))$, where $l_{\max}$ and $l_{\min}$ represent the maximum and minimum values of the profit and loss, and ϵ is the desired precision. In each iteration, we use the maximum likelihood estimation method to measure the target state, reducing the consumption of quantum resources. This optimization enhances the overall efficiency of the algorithm.

Algorithm 1: The Binary Search Algorithm for Searching Option VaR.

Input: confidence level α,the minimum value of option profit and loss level $l_{\min}$,the maximum value of option profit and loss $l_{\max}$
Output: option VaR l_α

```
repeat
    Initialize the profit and loss level l = (l_min + l_max)/2
    Constructing a target state |φ⟩_{n+1}|0⟩ less than l based on 𝒱
    Measuring the probability p̃_l of the (n+2)-nd quantum bit as |1⟩ in the state |φ⟩_{n+1}|0⟩
    if p̃_l < α then
        l_min = l
    else
        l_max = l
    end if
until l_min ≥ l_max
return l_α = l_min
```

Algorithm 1 illustrates the pseudocode of the binary search option VaR. Among them, we take the confidence level α, and the option profit and loss minimum $l_{\min}$ and the maximum $l_{\max}$ as input. The output is the VaR of the option at the final α level.

V. Experiment and Discussion

In this section, we select the common financial derivatives in the option market-European call options as the research object, use Qiskit [41] for simulation demonstration, and analyze the circuit depth of the model in detail and in-depth.

A. Experiment Setting

We consider a European call option with an initial price $S_0 = 2.0$, volatility $\sigma = 0.4$, expiration time of 40 days $T = 40/365$, and risk-free return rate $r = 0.05$. In the experiment, we tested different values of n, taking $n = 3$ as an example for detailed explanation.

We truncate the option price distribution to the interval within three standard deviations on both sides of the mean, and use three qubits $n = 3$ for encoding to load the probability distribution of the option prices. The qubits naturally represent integers through binary encoding. Due to the uniform sampling of the underlying asset prices, the fitting relationship between i and S_T can be expressed as $S_T = 0.229 \cdot i + 1.209$, where $i \in \{0, 1, \ldots, 7\}$ corresponds one-to-one with the possible values of S_T.

Next, we set the strike price of the option as $K = 1.896$. Using the fitted relationship, the corresponding integer for the strike price is calculated as $i_K = 3$. Thus, $k = 2$ qubits are required to encode the integer 3. Fig. 5 shows the result of loading the European call option distribution using three qubits, along with the corresponding payoff. After applying the operation $\mathcal{P}$, the resulting payoff distribution of the option is shown in Fig. 6. In this payoff distribution, the fitting relationship between the

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TABLE I
PARAMETER SETTINGS OF TWO OPTION INSTANCES

Option number j	Initial option price S_0	Volatility σ	Risk-free interest rate r	Expiration time T
1	2.0	0.4	0.05	40/365
2	1.8	0.5	0.06	55/365

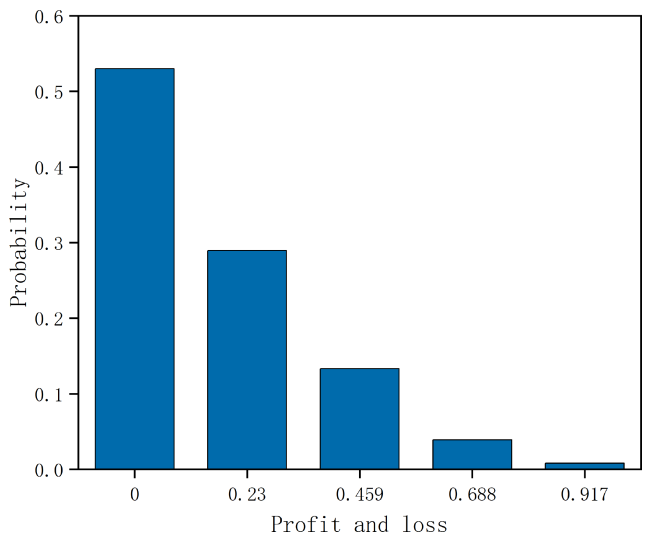

Fig. 6. The probability distribution diagram of PnL of option. The definition of the option is consistent with Fig. 5, and the exercise price is $K = 1.896$.

payoff value P_T and i can be expressed as:

$$P_T = \begin{cases} 0.229 \cdot i - 0.687, & i > i_K \\ 0, & i \leq i_K \end{cases} \tag{12}$$

After obtaining the option payoff distribution, we use binary search combined with (9) to estimate the Value at Risk at a confidence level of α, where α is set to 0.95. For the option risk estimation using Maximum Likelihood Estimation based amplitude estimation, an exponentially increasing sequence is adopted as the amplification sequence [39], where the sequence grows as follows: $f_0 = 0$, $f_1 = 2^0$, and so on, until $f_m = 2^{m-1}$. Here, we set $m = 6$, resulting in $f_q = \{0, 1, 2, 4, 8, 16, 32\}$. In addition, the total number of queries is set to $N_q = 100$.

To demonstrate the reliability and accuracy of the model in the face of different option distributions, we conducted simulations of two sets of European call options with different parameters. We evaluated the risk values at a confidence level of $\alpha = 0.95$ and used different numbers of sampling qubits n to load the option distributions. The specific parameters of the option are shown in Table I, with the exercise prices set at $K = 1.896$ and $K = 1.5$, respectively. In order to reduce the uncertainty introduced by the error in the measurement process and increase the credibility of the results, we measure the probability under each l value three times and then use the average of the three times as the probability sum of the profit and loss value less than l, that is $\mathbb{P}[P_T \leq l]$.

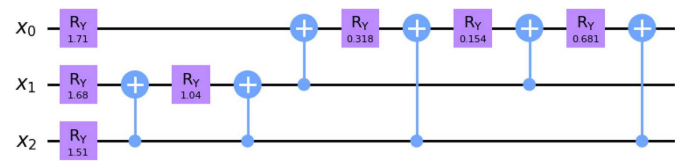

Fig. 7. The $\mathcal{L}$ operation of the considered option. This option has the same parameters as Fig. 5. x_0, x_1, x_2 represent quantum bits used to encode the option distribution.

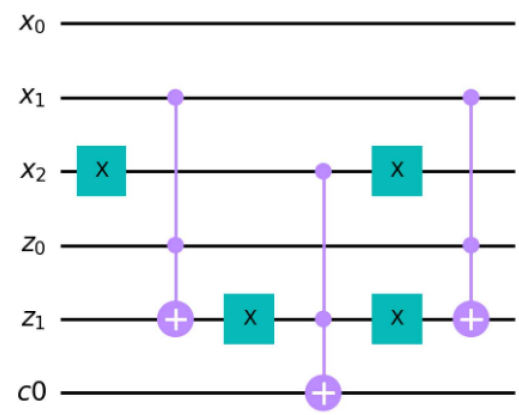

Fig. 8. The $\mathcal{P}$ operation of the considered option. The option price is compared with the given exercise price K, where $K = 1.896$. The qubit c_0 is a sign of whether the condition is satisfied, and the qubits z_0, z_1 represent the value to be compared K, which is clean at the beginning and end of the circuit.

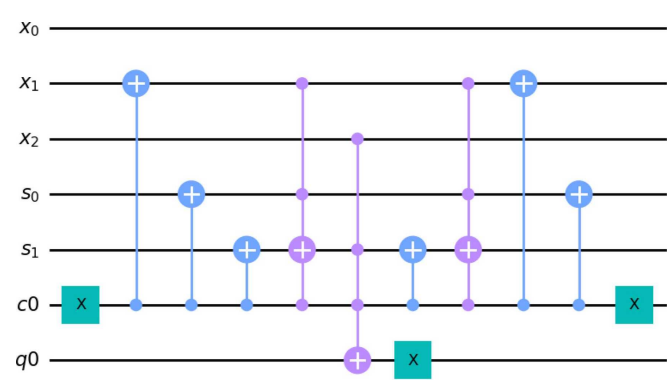

Fig. 9. The $\mathcal{V}$ operation of the considered option. Compare the PnL of option with a given l, here $l = 0.46$. The quantum bit q_0 is used to mark the state of whether the condition is satisfied. The qubits s_0, s_1 represent the value to be compared$l$. The circuit is clean both at the beginning and at the end.

B. Analysis and Discussion

Next, we conduct an analysis based on the component modules of the model. The QORA model mainly consists of the QOPLE algorithm and amplitude estimation based on maximum

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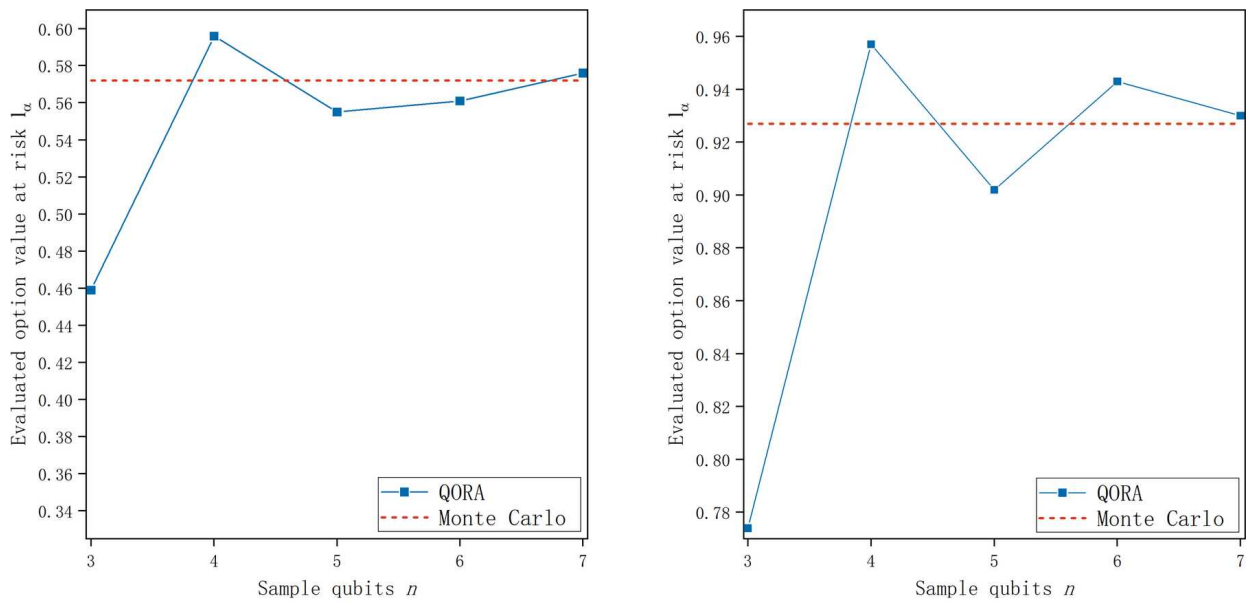

(a) Example One Option (b) Example Two Option

Fig. 10. In the figure, (a) and (b) correspond to the calculation results of the two options in the Table I, respectively. The blue solid line shows the VaR calculated by the QORA model when different qubits are loaded. The red dashed line represents the VaR calculated through classical Monte Carlo simulation with 10,000 repeated samples.

likelihood estimation. The QOPLE operation involves the operators $\mathcal{L}$, $\mathcal{P}$, and $\mathcal{V}$. Figs. 7, 8, and 9 show the circuit diagrams corresponding to the three operations, respectively.

We first use the operator $\mathcal{L}$ to load the probability distribution of the option. This operator is constructed through a series of unitary rotations and controlled-NOT gates, achieving the loading of the option price distribution into a quantum state represented by n qubits, with circuit complexity of $O(2^n)$. As n increases, the loaded model can provide more accurate distribution estimates.

This result aligns with our expectations, as increasing the number of sampling qubits allows us to accurately simulate the underlying log-normal distribution of the original data, thereby improving the accuracy of risk estimation. Consequently, the computed results become more precise. In both instances, for a sampling of $n = 7$ qubits, the relative error between the quantum algorithm and the classical algorithm is within 1%. This indicates that by appropriately selecting the value of n, the VaR obtained through quantum computation is within an acceptable range of error compared to those computed using classical algorithms.

The classical Monte Carlo method typically requires $O(1/\epsilon^2)$ samples to estimate the VaR value, where represents the computational precision. Under high precision requirements, this complexity increases significantly, leading to a sharp rise in computational time. In contrast, the QOPLE algorithm, leveraging Quantum Amplitude Estimation (QAE) techniques, reduces the complexity to $O(1/\epsilon)$, theoretically achieving quadratic speedup. This acceleration offers potential computational advantages for option risk measurement, allowing the quantum algorithm to perform risk estimation more efficiently under the same precision requirements.

Fig. 10 shows the VaR estimation results of two sets of options using the QOPLE algorithm at different sampling qubit numbers n, as well as the relative error between the quantum algorithm and the classical algorithm results (Relative Error = | Quantum Algorithm Result − Classical Monte Carlo Result | / Classical Monte Carlo Result × 100 %). It can be observed that as the number of sampling qubits n increases, the VaR estimates from the QOPLE algorithm gradually converge to those computed by the classical algorithm. For both sets of examples, when $n = 7$ sampling qubits are used, the relative error between the quantum and classical algorithms is within 1%. This suggests that by appropriately setting n, the VaR value obtained from quantum computation can be close to the value calculated by the classical algorithm, with the error remaining within an acceptable range. As the number of sampling qubits n increases, the quantum algorithm is able to more accurately model the option's probability distribution, reducing the error caused by insufficient sampling and thus improving the precision of the VaR estimation.

The operator $\mathcal{P}$ essentially compares the value in the X register with the integer value corresponding to the execution price encoded in the Z register. Because the value of the strike price K would be less than the maximum value of the option's expiration price, the integer values corresponding to K are less than or equal to $2^n - 1$. In other words, the Z register requires at most n bits, along with an auxiliary quantum bit for representing the comparison result. The operator $\mathcal{P}$ requires a maximum of $2n + 1$ quantum bits as inputs.

The operation $\mathcal{V}$ is essentially a controlled comparison circuit, where the auxiliary qubit $|c\rangle$ in $\mathcal{P}$ serves as the control qubit for the comparison. Since the range of risk values is limited by the payoff of the option, a maximum of n qubits is required to encode l in the S register when performing a binary

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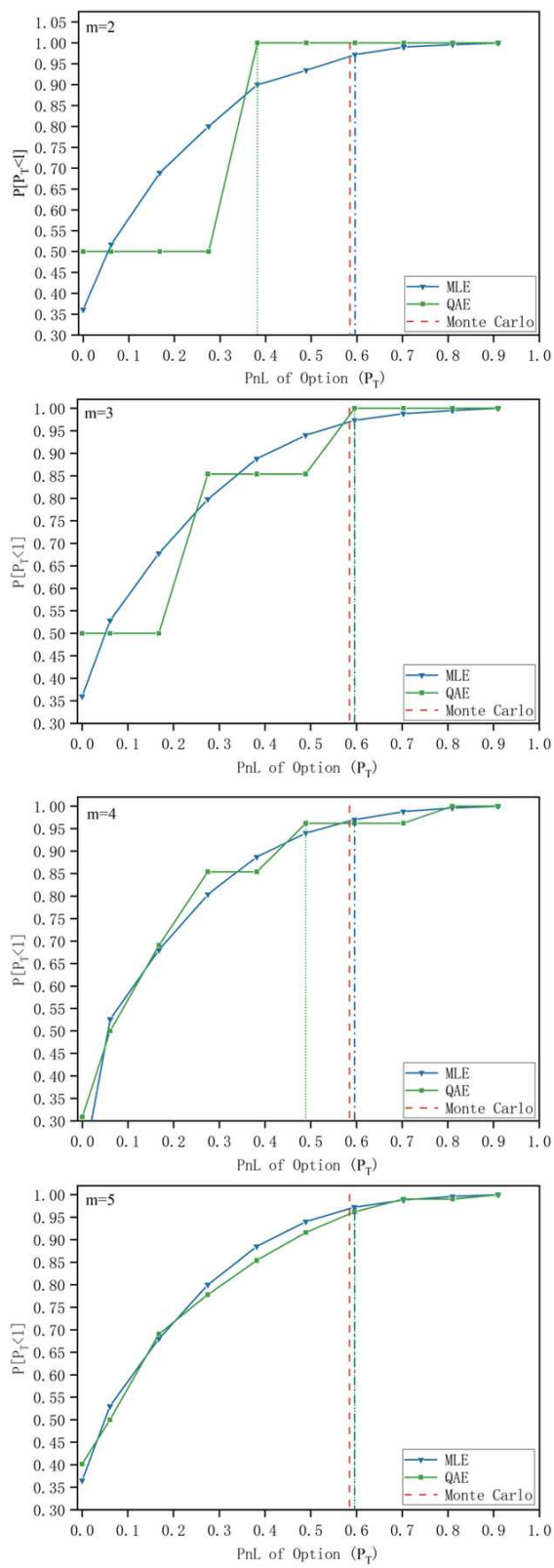

Fig. 11. Comparison of VaR estimated by QAE algorithm and MLE algorithm. The option parameter definition is consistent with Fig. 5, and it is discretized to four qubits $n = 4$. We show the evaluation results of the cumulative distribution function of the option profit and loss value P_T under the QAE algorithm (green) and the MLE algorithm (blue under $m \in \{2, 3, 4, 5\}$ and the VaR (green dotted line and blue dotted line) at the α level estimated by the final dichotomy. The red dotted line is the risk value calculated by 10000 times of sampling using classical Monte Carlo.

search to iteratively adjust l and compare it with the option payoff. Consequently, the operator $\mathcal{V}$ requires up to $3n + 2$ qubits as input. In practical applications, although higher option and strike prices increase the risk value and, subsequently, the qubit demand, the relationship among the strike price K, the given payoff level l, and the option maturity price S_T satisfies $S_{T_{\min}} \leq K \leq l \leq S_{T_{\max}}$. This ensures that the actual increase in qubit demand typically does not reach the theoretical upper bound. Moreover, the model exhibits strong scalability. In real market conditions, even when option and strike prices are higher, the model can adapt to more complex scenarios by increasing the number of qubits required for encoding, ensuring stability and effectiveness.

The query complexity of the estimated parameter is calculated by the number of operations of $\mathcal{A}$, which is expressed as the number of queries simplicity [39]. In the estimation of option risk value using maximum likelihood estimation, the total number of queries required for the sequence $\{f_q, N_q\}$ is computed as $N_t = n \cdot \sum_{q=0}^{R} N_q(2f_q + 1)$. One query is dedicated to preparing the initial state $|\phi\rangle_{n+1}|0\rangle$, $2f_q$ means that two $\mathcal{A}$ operations (i.e.$\mathcal{A}$ and $\mathcal{A}^{-1}$) are required for each amplitude amplification operation, and for the binary search, everything is repeated up to n times to estimate the risk value of the option.

The relationship between the accuracy ϵ of the classical method for estimating the probability of the target state and the sample size M_c can be expressed as $\epsilon = O(\frac{1}{\sqrt{M_c}})$. This means that to achieve accuracy ϵ, the required sample size is $M_c = O(\frac{1}{\epsilon^2})$. On the other hand, for the QAE method, the error ϵ and the quantum sample size M_q are related to $\epsilon = O(\frac{1}{M_q})$. Therefore, the sample size required to achieve the same precision is $M_q = O(\frac{1}{\epsilon})$. This indicates that QAE achieves a quadratic speed-up in sample complexity compared to classical methods. Specifically, for the same accuracy, the quantum sample size M_q is only the square root of the classical sample size M_c[19]. When using quantum amplitude estimation based on Maximum Likelihood Estimation (MLE), the amplification of the target state still follows the grover amplification mechanism, with the amplitude of the target state increasing non-linearly as $sin((2k + 1)\theta)$. The introduction of MLE mainly serves to optimize the accuracy of the estimation, rather than alter the measurement and amplification process. Therefore, even with MLE for estimation, the error convergence rate of the quantum algorithm still follows the pattern $O(\frac{1}{M_q})$. MLE optimizes the measurement results and post-processing, allowing for higher-precision amplitude estimation with fewer samples, thus reducing the number of measurements and improving efficiency. However, this optimization is based on the existing amplification mechanism and quantum interference of QAE, so the core source of the quadratic speedup is still preserved. The QORA model replaces phase estimation-based amplitude estimation with maximum likelihood estimation, as the former parameter estimation depends on auxiliary qubits and has high circuit depth resource requirements, making it more challenging to implement in practical applications [42].

Here, we present a simple example comparing the traditional QAE algorithm with the algorithm employed in our study. The simulation results are shown in Fig. 11. When $m = 2$, for the MLE algorithm, the amplification sequence is $\{f_q\} = \{0, 1, 2\}$.

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TABLE II
THE RESOURCE COMPARISON BETWEEN QAE ALGORITHM AND MLE ALGORITHM

Algorithm	The required qubits	The number of QFT	The number of Q	The number of controlled Q	Error
QAE	$3n+2+m$	1	-	2^m-1	$O(M^{-1})$
MLE	$3n+2$	0	2^m-1	-	$O(M^{-1})$

In this situation, the algorithm can estimate the relevant parameters more accurately according to the sample data, and further increasing the number of amplitude amplifications while not significantly improve the accuracy of the estimation.

In QAE algorithm, the goal is to estimate the amplitude $a = \sin^2\theta$ of a quantum state. However, the estimated value $\tilde{a}$ obtained through quantum circuits may deviate from the true value a. This deviation arises from factors such as the randomness of quantum measurements, the limitation of the number of qubits, and the uncertainty of quantum operations. The accuracy of the QAE algorithm depends on the number of qubits m and the number of quantum operations. When m is small, the estimated value fluctuates significantly, resulting in notable errors. As m increases, the accuracy of the estimation improves, and the error decreases. In contrast, the MLE algorithm exhibits smaller errors, accurately estimating parameters with fewer sample data, and the errors do not fluctuate significantly as the sample size increases. Although the errors of both QAE and MLE eventually converge to $O(1/M)$, their convergence behaviors differ. QAE shows larger error fluctuations when the sample size is small, whereas MLE achieves smaller and smoother errors under the same conditions. Beyond error performance, our MLE algorithm also demonstrates significant advantages in the utilization of quantum resources. Suppose the QAE algorithm requires m additional auxiliary qubits and defines the total number of Q-operator applications as $M = 2^{m-1}$. Since the MLE algorithm uses an exponential sequence as the amplification sequence, its general term can also be expressed as 2^{m-1}. Therefore, the quantum resources required for the both algorithms are shown in Table II. In this case, adopting the MLE algorithm reduces auxiliary qubits by at least 15% and decreases controlled amplification operations by three times. This improvement not only lowers hardware requirements and potential error rates but also enhances the accuracy and stability of practical quantum computations. This makes our algorithm more suitable for leveraging NISQ devices to achieve quantum advantage, particularly in the field of quantum finance, where it holds significant research and planning value.

VI. CONCLUSION

We introduce the quantum option profit and loss encoding (QOPLE) algorithm and successfully design the corresponding quantum circuit implementation. The algorithm focuses on the nonlinear characteristics of the payment structure of financial derivatives and realizes the loading of the option profit and loss into the quantum circuit. On the basis of QOPLE, we also construct the application model of the algorithm in the field of risk assessment, namely the quantum option risk assessment (QORA) model. The QORA model shows a significant quadratic acceleration advantage compared to the traditional classical Monte Carlo simulation. In addition, the model reduces the required quantum bits and the number of controlled operations by using maximum likelihood estimation, thereby reducing hardware requirements and potential error rates. This kind of resource optimization design improves the accuracy and stability of achievable quantum computing and makes us better adapt to the current level of quantum technology development.

Through simulation experiments on Qiskit, we validated the superior performance of the proposed quantum algorithm in option risk assessment, demonstrating its broad application potential and competitive advantages in this field. When dealing with massive data in the financial sector, this model leverages dimensionality reduction and the advantages of quantum computing to more efficiently solve option risk estimation problems, providing an effective tool for financial risk management and reducing the likelihood of potential risk events. It is worth noting that our quantum algorithm can provide a novel tool for the financial field. Future research can further expand the application of quantum algorithms in option risk assessment and explore broader financial scenarios and complex option contracts.

REFERENCES

[1] L. Zhang, X. Wu, H. Zhao, F. Cheng, and Q. Liu, "Personalized recommendation in P2P lending based on risk-return management: A multi-objective perspective," *IEEE Trans. Big Data*, vol. 8, no. 4, pp. 1141–1154, Aug. 2020.

[2] F. Wen, L. Chu, R. Ying, and P. Liu, "Fast and positive definite estimation of large covariance matrix for high-dimensional data analysis," *IEEE Trans. Big Data*, vol. 7, no. 3, pp. 603–609, Jul. 2021.

[3] A. Krause and S. Giansante, "Network-based computational techniques to determine the risk drivers of bank failures during a systemic banking crisis," *IEEE Trans. Emerg. Topics Comput. Intell.*, vol. 2, no. 3, pp. 174–184, Jun. 2018.

[4] W. Shi, W. -N. Chen, T. Gu, H. Jin, and J. Zhang, "Handling uncertainty in financial decision making: A clustering estimation of distribution algorithm with simplified simulation," *IEEE Trans. Emerg. Topics Comput. Intell.*, vol. 5, no. 1, pp. 42–56, Feb. 2021.

[5] D. Liu, J. Lv, and Y. Li, "Generating style-specific Chinese tang poetry with a simple actor-critic model," *IEEE Trans. Emerg. Topics Comput. Intell.*, vol. 3, no. 4, pp. 313–321, Aug. 2019.

[6] A. Gómez et al., "A survey on quantum computational finance for derivatives pricing and VaR," *Arch. Comput. Methods Eng.*, vol. 29, no. 6, pp. 4137–4163, Oct. 2022.

[7] M. Incudini, D. L. Bosco, F. Martini, M. Grossi, G. Serra, and A. Di Pierro, "Automatic and effective discovery of quantum kernels," *IEEE Trans. Emerg. Topics Comput. Intell.*, early access, Dec. 23, 2024, doi: 10.1109/TETCI.2024.3499993.

[8] Y. Koshka and M. A. Novotny, "Comparison of use of a 2000 Qubit D-wave quantum annealer and MCMC for sampling, image reconstruction, and classification," *IEEE Trans. Emerg. Topics Comput. Intell.*, vol. 5, no. 1, pp. 119–129, Feb. 2021.

[9] M. A. Nielsen and I. L. Chuang, *Quantum Computation Quantum Information: 10th Anniversary Edition*. New York, NY, USA: Cambridge Univ. Press, 2011.

[10] V. Dixit et al., "Training a quantum annealing based restricted Boltzmann machine on cybersecurity data," *IEEE Trans. Emerg. Topics Comput. Intell.*, vol. 6, no. 3, pp. 417–428, Jun. 2022.

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[11] H. Yu and X. Zhao, "Event-based deep reinforcement learning for quantum control," *IEEE Trans. Emerg. Topics Comput. Intell.*, vol. 8, no. 1, pp. 548–562, Feb. 2024.
[12] X. Pan et al., "Experimental quantum end-to-end learning on a superconducting processor," *NPJ Quantum Inf.*, vol. 9, no. 1, 2023, Art. no. 18.
[13] B. Bauer, S. Bravyi, M. Motta, and G. K.-L. Chan, "Quantum algorithms for quantum chemistry and quantum materials science," *Chem. Rev.*, vol. 120, no. 22, pp. 12685–12717, 2020.
[14] X. Li, M. Guo, Z. Wang, J. Li, and C. Qin, "Robust image hashing in encrypted domain," *IEEE Trans. Emerg. Topics Comput. Intell.*, vol. 8, no. 1, pp. 670–683, Feb. 2024.
[15] R. -H. Shi and Y. -F. Li, "Quantum secret permutating protocol," *IEEE Trans. Comput.*, vol. 72, no. 5, pp. 1223–1235, May 2023.
[16] D. Herman et al., "A survey of quantum computing for finance," 2022, *arXiv:2201.02773*.
[17] Y. -H. Chou, S. -Y. Kuo, and Y. -C. Jiang, "A novel portfolio optimization model based on trend ratio and evolutionary computation," *IEEE Trans. Emerg. Topics Comput. Intell.*, vol. 3, no. 4, pp. 337–350, Aug. 2019.
[18] H. Tang, A. Pal, T. -Y. Wang, L. -F. Qiao, J. Gao, and X. -M. Jin, "Quantum computation for pricing the collateralized debt obligations," *Quantum Eng.*, vol. 3, no. 4, Dec. 2021, Art. no. e84.
[19] D. J. Egger, R. García Gutiérrez, J. C. Mestre, and S. Woerner, "Credit risk analysis using quantum computers," *IEEE Trans. Comput.*, vol. 70, no. 12, pp. 2136–2145, Dec. 2021.
[20] Y. Ding et al., "Toward prediction of financial crashes with a D-Wave quantum annealer," *Entropy*, vol. 25, no. 2, p. 323, 2023, doi: 10.3390/e25020323.
[21] T. He, "Option pricing for path-dependent options with assets exposed to multiple defaults risk," *Discrete Dyn. Nature Soc.*, vol. 2020, Feb. 2020, Art. no. 2418620.
[22] T. Kariya and R. Y. Liu, "Options, futures and other derivatives," in *Asset Pricing: Discrete Time Approach.* Boston, MA, USA: Springer, 2003, pp. 9–26.
[23] O. Romanko and H. Mausser, "Robust scenario-based value-at-risk optimization," *Ann. Operations Res.*, vol. 237, no. 1/2, pp. 203–218, Feb. 2016.
[24] S. Mitra, "The relationship between conditional value at risk and option prices with a closed-form solution," *Eur. J. Finance*, vol. 21, no. 5, pp. 400–425, 2015.
[25] S. Mitra, "Efficient option risk measurement with reduced model risk," *Insurance, Math. Econ.*, vol. 72, pp. 163–174, Jan. 2017.
[26] E. Dri et al., "A more general quantum credit risk analysis framework," *Entropy*, vol. 25, no. 4, Mar. 2023, Art. no. 593.
[27] G. Brassard, P. Høyer, M. Mosca, and A. Tapp, "Quantum amplitude amplification and estimation," *AMS Contemporary Math*, vol. 305, pp. 53–74, 2002.
[28] A. Montanaro, "Quantum speedup of Monte Carlo methods," *Proc. Roy. Soc. A, Math., Phys. Eng. Sci.*, vol. 471, no. 2181, 2015, Art. no. 20150301.
[29] S. Woerner and D. J. Egger, "Quantum risk analysis," *NPJ Quantum Inf.*, vol. 5, no. 1, Feb. 2019, Art. no. 15.
[30] R. De Jongh, T. Verster, E. Reynolds, M. Joubert, and H. Raubenheimer, "A critical review of the Basel margin of conservatism requirement in a retail credit context," *Int. Bus. Econ. Res. J.*, vol. 16, no. 4, pp. 257–274, Oct. 2017.
[31] E. Dri, E. Giusto, A. Aita, and B. Montrucchio, "Towards practical quantum credit risk analysis," *J. Phys., Conf. Ser.*, vol. 2416, 2022, Art. no. 012002.
[32] S. Wilkens and J. Moorhouse, "Quantum computing for financial risk measurement," *Quantum Inf. Process.*, vol. 22, no. 1, 2023, Art. no. 51.
[33] M. Ammann and C. Reich, "VaR for nonlinear financial instruments linear approximation or full Monte Carlo?," *Financial Markets Portfolio Manage.*, vol. 15, no. 3, pp. 363–378, Sep. 2001.
[34] N. Stamatopoulos et al., "Option pricing using quantum computers," *Quantum*, vol. 4, 2020, Art. no. 291.
[35] P. Rebentrost, B. Gupt, and T. R. Bromley, "Quantum computational finance: Monte Carlo pricing of financial derivatives," *Phys. Rev. A*, vol. 98, no. 2, Aug. 2018, Art. no. 022321.
[36] P. Zhang, K. Yu, J. J. Yu, and S. U. Khan, "QuantCloud: Big data infrastructure for quantitative finance on the cloud," *IEEE Trans. Big Data*, vol. 4, no. 3, pp. 368–380, Sep. 2018.
[37] S. A. Cuccaro, T. G. Draper, S. A. Kutin, and D. P. Moulton, "A new quantum ripple-carry addition circuit," 2004, *arXiv:quant-ph/0410184*.
[38] H. Li et al., "Selective transfer based evolutionary multitasking optimization for change detection," *IEEE Trans. Emerg. Topics Comput. Intell.*, vol. 8, no. 3, pp. 2197–2212, Jun. 2024.
[39] Y. Suzuki, S. Uno, R. Raymond, T. Tanaka, T. Onodera, and N. Yamamoto, "Amplitude estimation without phase estimation," *Quantum Inf. Process.*, vol. 19, no. 2, Feb. 2020, Art. no. 75.
[40] A. Y. Kitaev, "Quantum measurements and the abelian stabilizer problem," 1995, *arXiv:quant-ph/9511026*.
[41] G. Aleksandrowicz et al., "Qiskit: An open-source framework for quantum computing," 2019. [Online]. Available: https://github.com/Qiskit
[42] D. Grinko, J. Gacon, C. Zoufal, and S. Woerner, "Iterative quantum amplitude estimation," *NPJ Quantum Inf.*, vol. 7, no. 1, Mar. 2021, Art. no. 52.

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